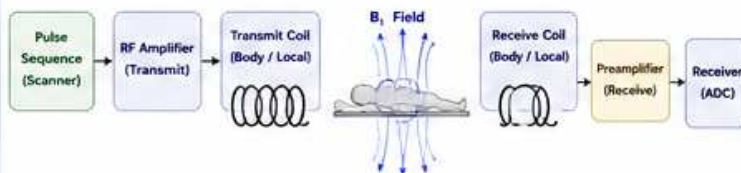


RF problems show up in the image. Recognize the pattern, confirm with measurements, and fix the RF chain.

Recognize (Image Clues) → Isolate (Tests) → **Fix (Root Cause)** → Verify (Image & Metrics)

## 1. THE BIG PICTURE – RF SIGNAL PATH



- Any problem in the RF chain can cause poor or wrong images.
- Use image clues + simple tests to localize the fault.

## 2. COMMON RF PROBLEMS & WHAT THEY LOOK LIKE

| Problem                              | Typical Image Appearance                   |                                                                                     | What You May See                            | Suspect Area                                   |
|--------------------------------------|--------------------------------------------|-------------------------------------------------------------------------------------|---------------------------------------------|------------------------------------------------|
| Low Transmit Power or Bad Tx Coil    | Low signal, low SNR, poor fat sat          |  | Overall dark image, noise may be normal     | TX amp, Tx coil, connectors, detune            |
| Overdriven / High Transmit Power     | Burned center, shading, non-uniform signal |  | Bright center, signal drop at edges         | Tx gain too high, B <sub>1</sub> inhomogeneity |
| Tx/Rx Coil Mismatch or Bad Tuning    | Low SNR, poor uniformity, banding          |  | Low signal, standing wave patterns          | Coil tuning/match, cables, traps               |
| Detuned Coil (Transmit Not On)       | Almost no signal, very noisy               |  | Noise only or near zero signal              | Diodes, detune circuit, cable                  |
| Bad Preamplifier or Cable            | High noise, streaks, fixed pattern         |  | Vertical/horz lines, spikes, noise floor up | Preamplifier, cable, connections               |
| Partial Element Failure (Array Coil) | Signal voids or asymmetric areas           |  | Missing signal in one area/side             | Element, cable, preamp channel                 |

## 3. QUICK IMAGE CLUES (WHAT TO LOOK FOR)

|                                |                                                                                                         |
|--------------------------------|---------------------------------------------------------------------------------------------------------|
| <b>Uniform Low Signal</b>      | <br>• Everything dark<br>• Noise level normal<br>→ Think: Low Tx power, detune, bad Tx path             |
| <b>Shading / Bright Center</b> | <br>• Center too bright<br>• Signal drops at edges<br>→ Think: Overdrive, B <sub>1</sub> inhomogeneity  |
| <b>Banding / Stripes</b>       | <br>• Dark/light bands<br>• Change with FOV/position<br>→ Think: Mismatch, standing waves, cable issues |
| <b>Noise Only / No Signal</b>  | <br>• No anatomy<br>• Just noise<br>→ Think: Detuned coil, Tx off, open circuit                         |
| <b>Streaks / Spikes</b>        | <br>• Lines or spikes<br>• Worse on one axis<br>Cable, EMI                                              |
| <b>Localized Signal Void</b>   | <br>• Missing signal in one area<br>Normal elsewhere<br>→ Think: Element failure, Connection issue      |

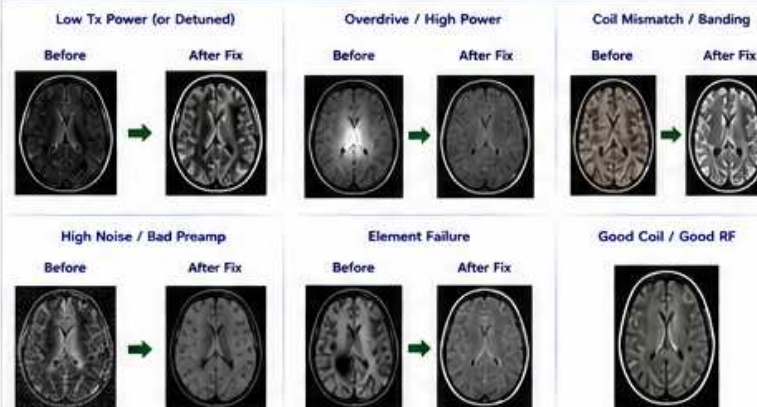
## 4. ISOLATE THE PROBLEM (TESTS)

- Check Scanner Status**  
Any RF amp faults? Interlocks?  
Verify Tx/Rx channels OK.
- Run Prescan / Coil Check**  
Look at noise (σ), PNS, and coil status.  
Compare to baseline.
- Swap Coils**  
If available, swap with known good coil.  
Problem follows coil = coil issue.
- Cable Wiggle Test**  
Move cables gently at coil and preamp.  
If image changes → bad connection.
- B<sub>1</sub> Mapping (Transmit)**  
Check for low B<sub>1</sub> or severe inhomogeneity.  
Confirms Tx chain issues.
- Noise Only Scan (No RF)**  
Run with RF off. If noise is high → receiver chain issue.

## 5. COMMON ROOT CAUSES & FIXES

| Root Cause                | Why It Happens                        | What to Check                                  | Fix                                        |
|---------------------------|---------------------------------------|------------------------------------------------|--------------------------------------------|
| Low Tx Power / Amp Issue  | Amp fault, low gain, interlock        | Amp status, forward/reflected power, temp      | Reset, service amp, check cooling          |
| Tx Coil Cable / Connector | Loose, open, or short                 | Visual, wiggle test, continuity                | Tighten, clean, or replace                 |
| Coil Detuned              | Detune failure, diode open            | Detune LED, voltages, Q with/without detune    | Replace diode, repair detune circuit       |
| Coil Mismatch             | Tuning drift, wrong cable, trap issue | Return Loss (S11), Smith chart                 | Re-tune, re-match, replace parts           |
| Overdrive / High Tx Gain  | Gain too high, wrong calibration      | B <sub>1</sub> map, power levels, SAR headroom | Reduce gain, re-calibrate, update protocol |
| Preamplifier / Cable      | Bad preamp, cable, EMI                | Noise scan, channel test, replace & compare    | Replace preamp, replace cable              |
| Array Element Failure     | Element, cable, or connector          | Element test, channel SNR compare              | Repair connection, replace element         |

## 6. BEFORE & AFTER EXAMPLES (REAL-WORLD PATTERNS)



The image tells you where to look. Tests confirm. The fix restores uniform, high-quality images.

## 7. RF TROUBLESHOOTING CHECKLIST (QUICK REFERENCE)

- Review symptoms and image pattern.
- Check system messages and RF status.
- Run prescan / coil check (noise, PNS).
- Compare with baseline or other coil.
- Check cables, connectors, and coil condition.
- Check tuning/matching (Return Loss).
- Check transmitter (forward/reflected power).
- Check B<sub>1</sub> map and uniformity.
- Test preamplifier and receiver chain.
- Document findings and verify after repair.

## 8. KEY TAKEAWAYS

- RF problems have characteristic image patterns.
- Start with the image, then confirm with simple tests.
- Isolate to Tx, Rx, coil, or cables.
- Fix the root cause, not the symptom.
- Always verify with images, metrics, and the patient.



### ENGINEER NOTES

- Keep baseline data: noise, PNS, Return Loss, B<sub>1</sub> maps.
- Small changes in tuning can cause big image problems.
- A good RF chain = good power + good match + low noise.

### ABBREVIATIONS

|                                               |                                |
|-----------------------------------------------|--------------------------------|
| B <sub>1</sub> = RF Magnetic Field (Transmit) | SAR = Specific Absorption Rate |
| SNR = Signal-to-Noise Ratio                   | Tx = Transmit                  |
| PNS = Prescan Normalize Signal                | Rx = Receive                   |



### SAFETY REMINDER

- High RF power can cause tissue heating.
- Follow all site safety procedures.
- Do not bypass interlocks.



### THE BOTTOM LINE

Recognize the image pattern.  
Run the right tests.  
Find the cause. Fix it. Verify it.  
Great RF performance = great patient images.



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